

Brady Hess

Physics Midterm 2

1. $1/1.33 \text{ sec}$

RPM = 45, 4.7 radians

(B)

2. $a_c = (4.7)^2 \times 0.10$

$a_c = 22.69 \times 0.10 = 2.21 \text{ m/s}^2$

(D)

3. $\frac{a_c}{a_t} = 100$

$\frac{w^2}{r} = \sqrt{100} = 10$

(D)

4. $w = 200 \times \frac{2\pi}{60} = \frac{400\pi}{60}$

$a_c = w^2 \cdot r = (20.94)^2 \times 0.12 = 438.5 \times 0.12 = 52.62$

$F_c = 0.01 \times 52.62 = 0.5262 \text{ N}$

(B)

5. $v = w \cdot r$

$v = 30.0 \text{ rad/s} \times 1.40 \text{ m}$

$v = 42.0 \text{ m/s}$

(C)

UNIT 5

1. $w = (40\hat{i} - 30\hat{j}) \cdot (5\hat{i}) = 200 \text{ N}$

$\sqrt{40^2 + (-30)^2} = \sqrt{2500} = 50 \text{ N}$

$\cos \theta = \frac{200}{50 \times 5} = \frac{200}{250} = 0.8$

$\theta = \cos^{-1}(0.8) \approx 36.87^\circ$

(D)

2. $w = F \cdot \Delta x = 1000 \text{ N} \times 1.0 \text{ m} = 1000 \text{ J}$

$A = 100 \text{ J}$

$\Delta KE = w$

$\Delta KE = 100 \text{ J}$

$B = 100 \text{ J}$

$100 = \frac{1}{2} \times 0.15 \times v^2$

$v^2 = \frac{200}{0.15} = 1333.33$

$C = 36.5 \text{ m/s}$

$R = \frac{v^2 \sin^2(2\theta)}{g}$

$R = \frac{(36.5)^2}{9.8} = \frac{1332.25}{9.8}$

$D = 136.5 \text{ m}$

3. $w = F \cdot d \cdot \cos \theta$

$N = m \cdot g = 40 \text{ kg} \times 9.8 \text{ m/s}^2 = 392 \text{ N}$

$N = 0.05 \times 392 \text{ N} = 19.6 \text{ N}$

$w = d \cdot 19.6 \times 40 = 784 \text{ J}$

$A = 784 \text{ J}$

$P = \frac{784 \text{ J}}{30} = 26.13 \text{ W}$

$B = 26.13 \text{ W}$

$$4.3W \text{ } 9c/W$$

$$P = \frac{3.00W}{1000} = 0.003kW$$

$$(8766) = 26.28$$

$$(0.09) = 2.3$$

(D)

UNIT 6

$$1. (20 \times 10^{-25} \text{ kg}) (350) (350) (20 \times 10^{-25})$$

$$2 \cdot m = 2 \times 20 \times 10^{-25} \text{ kg} = 40 \times 10^{-25}$$

$$G = 46 \times 10^{-25} \cdot V F$$

$$V F = 0 \text{ m/s}$$

(C)

$$2. x = V \cos(45) = 350 \times \cos(45) = 350 \times 0.707$$

$$y = 350 \times \sin(45) = 350 \times \frac{1}{2} = 217.45$$

$$P_y = m \cdot V_{iy} + m \cdot a_y = 20 \times 10^{-25} \cdot 0 + 20 \times 10^{-25} \cdot 247.45$$

$$= 20 \times 10^{-25} \cdot 247.45 = 4.949 \times 10^{-23}$$

$$V_F = \sqrt{298.725^2 + 123.725^2}$$

$$V_F = \sqrt{89285.6 + 15303.9} = \sqrt{104589.5}$$

$$(323.52 \text{ m/s})$$

$$3. F \cdot \Delta t = 4000 \text{ N} \cdot 0.200 \text{ s} = 800 \text{ N} \cdot \text{s}$$

$$A = 800 \text{ N}$$

$$800 = 200(V_F - 2.80)$$

$$V_F = 4 + 2.8 = 6.8 \text{ m/s}$$

$$B = 6.80 \text{ m/s}$$

$$4. C, \text{ elastic}$$

$$5. 30,000 \cdot .850 +$$

$$110,000 = 0$$

$$25,500 / 140,000 = 0.182$$

$$V = \frac{25,500}{140,000} = 0.182 \text{ m/s}$$

$$A = 0.182 \text{ m/s}$$

$$KE_{initial} = \frac{1}{2} m_{car} v_{car}^2$$

$$\frac{1}{2} (140,000) \cdot (.182) = \frac{1}{2} (140,000)$$

$$= 2,317.5$$

$$10,837.5 - 2,317.5$$

$$B = 8,520.5 \text{ J}$$

UNIT 7

$$1. F = 10 \cdot 9.8 = 392 \text{ N}$$

$$60 \cdot 9.8 = 588 \text{ N}$$

$$392.3 = 1176$$

$$588 \cdot 2 = 1176$$

equal

(A)

2. 2m plank

$$Mg = 20kg \times 9.8m/s^2 = 196.0N$$

$$F_1 + F_2 = 196N$$

$$= 196.0N \times 1.00m = 196.0N \cdot m$$

$$196.0N \cdot m = F_2 \times 1.50m$$

$$F_2 = \frac{196.0N \cdot m}{1.50m} = 130.67N$$

$$F_1 + 130.67N = 196.0N$$

$$F_1 = 65.33N$$

$$F_2 = 130.67N$$

UNIT 8:

$$1. I = \frac{1}{2} \cdot (1kg) \cdot (0.08m)^2$$

$$I = \frac{1}{2} \cdot 1(0.0664) = 0.0032kg \cdot m^2$$

$$= 0.0032kg \cdot m^2$$

$$T = (0.0032kg \cdot m^2)(2rad/s^2)$$

$$T = 0.0064N \cdot m$$

$$A = 0.0032kg \cdot m^2$$

$$B = 0.0064N \cdot m$$

$$2. T = \hat{i}(0) - \hat{j}(0) + k(-2)(0) + 12$$

$$T = k(-2.4)$$

$$T = -2.4kN \cdot m$$

$$T_{opp} = 2.4kN \cdot m$$

$$r \times F_{opp} = 2.4\hat{k}$$

$$(-.2)F_y - (.20)F_x = 2.4$$

$$-0.20F_y - 0.20F_x = 2.4$$

$$F_x + F_y = -12$$

$$A = 2.4kN \cdot m$$

$$B = 12N$$